TD 1

EXER 1

① At $T = 300K$

$B = 7{,}3 \times 10^{15} cm^{-3} K^{-3/2}$

$E_g = 1{,}2 eV$

$K = 8{,}62 \times 10^{-5} eV/K$

$n_i = BT^{3/2} e^{-E_g/2KT}$

$= 7{,}3 \times 10^{15} \times 300^{3/2} \times e^{-1{,}2/2\times 8{,}62\times 10^{-5}\times 300}$

$= 3{,}01 \times 10^{9}$ atoms/cm^3

At $T = 600K$

$n_i = 7{,}3 \times 10^{15} \times 600^{3/2} e^{-1{,}2/2\times 8{,}62\times 10^{-5}\times 600}$

$= 9{,}84 \times 10^{14}$ atoms/cm^3

∴ Because there is more energy to liberate the atoms from their bonds.

② pour $T = 300K$

$n_n = N_D = 10^{16}$ ~~atoms~~ electrons/cm^3

$P_n = \dfrac{n_i^2}{N_D} = \dfrac{(3{,}01\times 10^{9})^2}{10^{16}}$

$= 906{,}01$ holes/cm^3

EXERCICE 2

① $L = 1\,\mu m$, $V = 1V$, $\mu_n = 1350\ cm^2/Vs$

$E = \dfrac{V}{L} = \dfrac{1}{1\times 10^{-4}} = 10^4 V/cm$

$V_{n\text{-drift}} = -\mu_n E$

$= -(-10^4 \times 1350)$

$= 1{,}35 \times 10^7$ cm/s

②

We want to have

$I_n = I_p$

$\cancel{A}\cancel{q} n \mu_n E = \cancel{A}\cancel{q} P \mu_p E$

$n\mu_n = P\mu_p$

$\dfrac{n}{P} = \dfrac{\mu_p}{\mu_n}$

$\dfrac{n}{P} = \dfrac{480}{1350} = 0{,}356$

$n = 0{,}356 P$

EXERCICE 3

$T = 300K$, $N_A = 10^{18}/cm^3$, $N_D = 10^{16}/cm^3$

$A = 10^{-4} cm^2$

① $P_p = N_A = 10^{18}/cm^3$

$n_{p0} = \dfrac{n_i^2}{N_A} = \dfrac{(1{,}5\times 10^{10})^2}{10^{18}}$

$= \underline{225/cm^3}$

$n_n = N_D = 10^{16}/cm^3$

$P_{n0} = \frac{n_i^2}{N_D} = \frac{(1,5\times10^{10})^2}{10^{16}} = 2,25\times10^4/cm^3$

$V_0 = V_T \ln\left(\frac{N_D \times N_A}{n_i^2}\right)$

$V_T = \frac{KT}{q} = \frac{8,62\times10^{-5}(eV/K)\times300(K)}{1e}$

$= 0,025V = 25mV$

$V_0 = 0,025 \ln\left(\frac{10^{18}\times10^{16}}{(1,5\times10^{10})^2}\right)$

$V_0 = 0,79\,V$

EXERCICE 4

ⓐ $i_D = K_n'\left(\frac{W}{L}\right)V_{ov}^2$ et $K_n' = C_{ox}\mu_n$

$V_{ov} = \sqrt{\frac{i_D.2.L}{C_{ox}.\mu_n.W}} = \sqrt{\frac{10^{..}.2.0,18}{8,6\times10^{-15}\times450.10^{8}.2}}$

$= 215,6\,mV$

$V_{ov} = V_{GS} - V_t =$

$V_{GS} = V_{ov} + V_t$

$V_{GS} = 215,6\times10^{-3} + 0,5$

$= 0,715V$

$V_{DS} = V_{ov} = 215,6\,mV$

ⓑ $i_D = K_n'\left(\frac{W}{L}\right)\left(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2\right)$

$K_n' = \mu_n C_{ox} = 450\times10^{-4}\times\frac{8,6\times10^{-15}}{10^{-12}}$

$= 450\times10^{-4}\times8,6\times10^{-3}$

$K_n' = 3,87\times10^{-4}A/V^2$

$W/L = \frac{2}{0,18} = 11,111$

$i_D = 50\mu A = 50\times10^{-6}A$

Alors

$50\times10^{-6} = 3,87\times10^{-4}\times11,111(0,215V_{DS} - 0,5V_{DS}^2)$

$0,0116 = 0,215V_{DS} - 0,5V_{DS}^2$

$-0,5V_{DS}^2 + 0,215V_{DS} - 0,0116 = 0$

$V_{DS} = 0,367V$, $V_{DS} = 0,0632V$

Donc $V_{DS} = 0,0632V$ car $V_{DS} < V_{ov} = 0,215V$

ⓒ $V_{DS} = 0,3V$, V_{GS_2}

$I_{D_1} = \frac{1}{2}K_n'\left(\frac{W}{L}\right)V_{ov}^2$

$= \frac{1}{2}\times3,87\times10^{-4}\times11,111\times0,2^2$

$I_{D_1} = 86\mu A$

Pour $V_{GS_2} = 0,7 + 0,01 = 0,71V$

$I_{D2} = 94,8\mu A$

$\Delta I_D = I_{D2} - I_{D1} = 8,8\mu A$

pour $V_{GS_3} = 0,7 - 0,01 = 0,690\ V$

$I_{D3} = 77,6\ \mu A$

$\Delta I_D = I_{D3} - I_{D1}$

$= -8,4\ \mu A$

EXERCICE 5

(a)

$L_3 < L_2 < L_1$

$$R = \frac{L}{K_n' W V_{OV}}$$

* lorsque L augmente, la résistance aussi augment. $L \uparrow \Rightarrow R \uparrow \Rightarrow I_D \downarrow (V_G > V_{TH})$

* pour $V_{GS} < V_{TH} \Rightarrow I_D = 0$

(b)

$$I_D = K_n \frac{W}{L}\left(V_{OV} V_{DS} - \frac{1}{2} V_{DS}^2\right)$$

(d)

(c) $W_3 > W_2 > W_1$

EXERCICE 6

* Assuming Saturation (i)

$V_{DD} = 1,8\ V$, $\frac{W}{L} = \frac{2}{0,18} = 11,111$, $\mu C_{ox} = 100 \times 10^{-6}\ A/V^2$

$V_{GS} = V_G - V_S = 1 - 0$

$V_{GS} = 1$ alors $V_{OV} = V_{GS} - V_{TH}$

$= 1 - 0,4 = 0,6$

$$I_D = \frac{1}{2} K_n' \frac{W}{L} V_{OV}^2$$

$$= \frac{1}{2} \times 100 \times 10^{-6} \times 11,111 \times 0,6^2$$

$I_D = 2 \times 10^{-4}\ A = 200\ \mu A$

* Vérification

$V_D = V_{DD} - I_D R_D = 1,8 - (5000 \times 2 \times 10^{-4})$

$V_D = 0,8\ V$

$V_{DS} = V_D - V_S = 0,8 - 0 = 0,8 > V_{OV}$

(ii)

* si $V_{GS} \uparrow \rightarrow V_{GS} = 1 + 0,01 = 1,01\ V$

$$I_D = \frac{1}{2} K_n' \frac{W}{L} [1,01 - 0,4]^2$$

$I_D = 206,7\ \mu A$

$V_D = V_{DD} - R_D I_D$

$= 1,8 - 5 \cdot 10^3 \times 206,7 \cdot 10^{-6}$

$V_D = 0,766\ V$

$\Delta V_D = 0,8 - 0,766 = 0,034\ V$

TD 2

EXERCICE 1

① $K'_n = C_{ox} \mu_n \qquad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$

Alors

$$K'_n = \frac{\varepsilon_{ox}}{t_{ox}} \cdot \mu_n$$

$$= \frac{3{,}45 \times 10^{-11}}{4 \times 10^{-9}} \times 450 \times 10^{-4}$$

$$K'_n = 3{,}88 \times 10^{-4} \, A/V^2$$

②

$$r_{DS} = \frac{1}{\mu_n C_{ox} W/L \, (V_{GS} - V_t)}$$

$$W = \frac{L}{K'_n \, r_{DS} (V_{GS} - V_t)}$$

$$W = 9{,}28 \times 10^{-7} m = 0{,}928 \, \mu m$$

③ $I_D = \frac{1}{2} K'_n \frac{W}{L} V_{ov}^2$

$$V_{ov} = \sqrt{\frac{2 I_d \times L}{K'_n W}} = \sqrt{\frac{2 \times 0{,}3 \times 10^{-3} \times 1}{3{,}88 \times 10^{-4} \times 20}}$$

$$V_{ov} = 0{,}27 V$$

④ V_{DS} needed is $0{,}27V$

EXERCICE 2

ⓐ $V_{GS} < V_{tp} \Rightarrow |V_{GS}| > |V_{tp}|$

$V_G - V_S < V_{tp}$

$V_G < V_{tp} + V_S$

$V_G < -1 + 5$

$V_G < 4V$

ⓑ In triode region

$V_{DG} > |V_{tp}|$

$V_D - V_G > |V_{tp}|$

$V_D > |V_{tp}| + V_G$

$V_D > V_G + 1$

ⓒ In saturation

$V_{DG} \leq |V_{tp}|$

$V_D \leq |V_{tp}| + V_G$

$V_D \leq 1 + V_G$

ⓓ $\lambda = 0$

(i) $I_d = \frac{1}{2} K'_p \frac{W}{L} |V_{ov}|^2$

$$|V_{ov}| = \sqrt{\frac{2 I_d}{\frac{W}{L} \cdot K'_p}} = \sqrt{\frac{2 \times 75}{10 \times 60}}$$

$$|V_{ov}| = 0{,}5 V$$

(ii) $|V_{ov}| = 0{,}5 = V_{SG} - |V_{tp}|$

$V_{SG} = 0{,}5 + |V_{tp}| = 1{,}5V$

$V_S - V_G = 1{,}5$

$V_G = -1{,}5 + V_S = 3{,}5V$

(iii) $V_D \leq 1 + V_G$

$V_D \leq 1 + 3{,}5$

$V_D \leq 4{,}5$

(e)

$$r_o = \frac{1}{|\lambda| I_D}$$

$$= \frac{1}{0{,}02 \times 75 \times 10^{-6}}$$

$$r_o = 667\ K\Omega$$

(f)

i) $I_D = \frac{1}{2} K'_p \left(\frac{W}{L}\right) V_{ov}^2 (1 + |\lambda| V_{SD})$

Si $V_{SD} = V_S - V_D = 5 - 3 = 2V$

$$I_D = \frac{1}{2} \times 60 \times 10^{-6} \times 10 \times 0{,}5^2 \times (1 + 0{,}02 \times 2)$$

$$I_D = 78\ \mu A$$

ii, pour $V_D = 0 \Rightarrow V_{SD} = 5V$

$$I_D = \frac{1}{2} \times 60 \times 10^{-6} \times 10 \times 0{,}5^2 \times (1 + 0{,}02 \times 5)$$

$$I_D = 82{,}5\ \mu A$$

iii)

$$r_{DS} = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{3}{4{,}5\ \mu A} = 667\ K\Omega$$

EXERCICE 3

(1)

$$R = \frac{V_{DD} - V_D}{I_D} \qquad R = \frac{V_{DD} - V_D}{I_D}$$

$$I_D = \frac{1}{2} K_n V_{ov}^2 \qquad = \frac{1{,}8 - 0{,}7}{0{,}032}$$

$$I_d = 0{,}032\ mA \qquad R = 34{,}4\ K\Omega$$

(2) At the edge of saturation

$V_{DS} = V_{ov}$ ~~$= V_{GS} - V_t$~~ $\quad V_{DS} = V_D - V_S$ ($V_S = 0$)

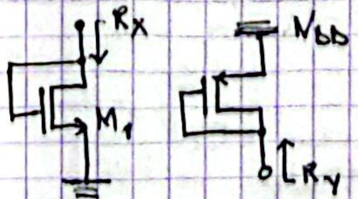
$$V_D = V_{GS} - V_t = 0{,}7 - 0{,}5 = 0{,}2V = V_{ov}$$

$$R_2 = \frac{V_{DD} - V_D}{I_D}$$

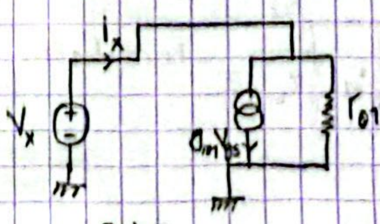
$$R_2 = \frac{1{,}8 - 0{,}2}{0{,}032 \times 10^{-3}}$$

EXERCICE 4

(a)



NMOS Connected-diode   PMOS



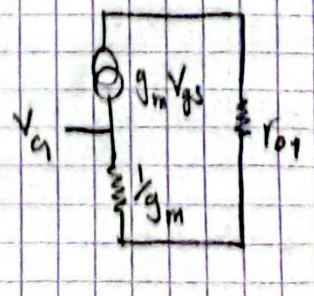
$$R_X = \frac{V_x}{i_x}$$

$\lambda \neq 0$

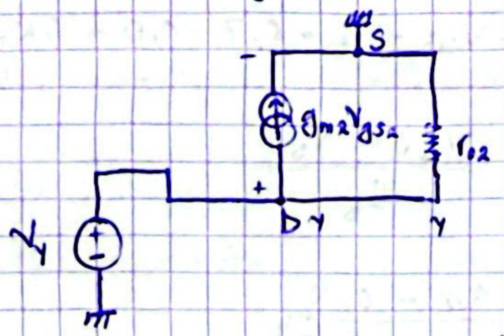
$$i_x = g_m V_{gs} + \frac{V_x}{r_{o1}} = g_{m1} V_x + \frac{V_x}{r_{o1}}$$

$$= \left(g_{m1} + \frac{1}{r_{o1}}\right) V_x$$

$$R_x = \frac{1}{g_{m1}} // r_{o1} \qquad R_1 // R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$



$$\left[r_{o1} // \frac{1}{g_{m1}}\right] = \frac{r_{o1}\cdot\frac{1}{g_{m1}}}{r_{o1}+\frac{1}{g_{m1}}}$$



$R_y = \frac{V_y}{I_y}$, $V_y = r_{o2}(I_y - g_{m2}V_{gs2})$

with $V_{gs2} = V_y$

$V_y(1+g_{m2}r_{o2}) = r_{o2} I_y$

$$R_y = \frac{V_y}{I_y} = \frac{r_{o2}}{1+g_{m2}r_{o2}} = r_{o2} // \frac{1}{g_{m2}}$$

EXERCICE 5

(1)

$\equiv$ $R'_{eq} = r_{o1}/R_D$

$A_v = \frac{V_{out}}{V_{in}}$; $V_{gs1} = V_{in}$

$V_{out} = -R'_{eq}\cdot g_{m1}V_{gs1} = -R'_{eq}\cdot g_{m1}V_{in}$

$A_v = -R'_{eq}\cdot g_{m1}$

* if $\lambda = 0$; $\Rightarrow R'_{eq} = R_D$. $\frac{1}{360\,\Omega}$

$A_v = -g_{m1}\cdot R_D$

$g_{m1} = \sqrt{2\cdot K_n I_D} = \sqrt{2K'_n \frac{W}{L} I_D}$

$= 3{,}33\cdot 10^{-3}\,\Omega^{-1}$

$A_v = -3{,}33\cdot 10^{-3}\cdot 10^{3} = -3{,}33$ V/V

(2)

* $V_{DS} = V_{DD} - R_D\cdot I_D$

$= 0{,}8$ V

* $V_{GS} = ?$

$I_D = \frac{1}{2}K'_n \frac{W}{L}(V_{GS}-V_t)^2$

$\Rightarrow V_{GS} = V_t + \sqrt{\frac{2I_D}{K'_n \frac{W}{L}}} = 1{,}1$ V

$\Rightarrow V_{ov} = V_{GS} - V_t = 0{,}6$ V

$\Rightarrow V_{DS} > V_{GS} - V_t \Rightarrow M_1$ operates in saturation region.

EXERCICE 6

(1)

$$V_{ov}|_B = \frac{\sqrt{2K_nR_DV_{DD}+1}-1}{K_nR_D}$$

$V_{ov}|_B = 0{,}213$ V

$V_{GS}|_B = V_t + V_{ov}|_B$

$= 0{,}4 + 0{,}213$

$V_{GS}|_B = 0{,}613$ V

* $V_{DS}|_B = V_{ov}|_B = 0{,}213$ V

(2) $V_{DS}|_C$?

$I_D = K_n\left[(V_{GS}|_C - V_t)V_{DS}|_C - \frac{V_{DS}|_C^2}{2}\right]$

$I_D = K_n(V_{GS}|_C - V_t)V_{DS}|_C$

$I_D = 4(1{,}8-0{,}4)V_{DS}|_C = 5{,}6\,V_{DS}|_C$ (mA)

* $I_D = \frac{V_{DD}-V_{DS}|_C}{R_D} \simeq \frac{V_{DD}}{R_D} = \frac{1{,}8}{17{,}5\cdot 10^3}$

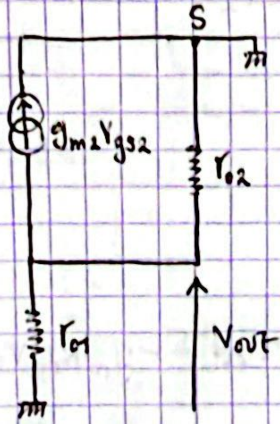
$= 0{,}1$ mA

$\therefore V_{DS}|_C = \frac{I_D}{5{,}6} = \frac{0{,}1\cdot 10^{-3}}{5{,}6} = 18$ mV

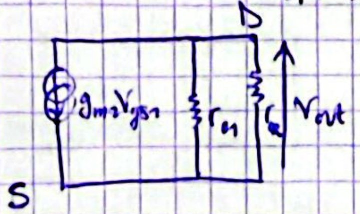
TD 3

EXERCICE 1

①



$R_{out} = r_{o1} // r_{o2} = \frac{r_{o1} \cdot r_{o2}}{r_{o1} + r_{o2}}$

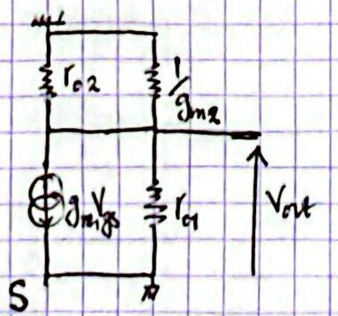


with $v_{gs} = V_{in}$

②

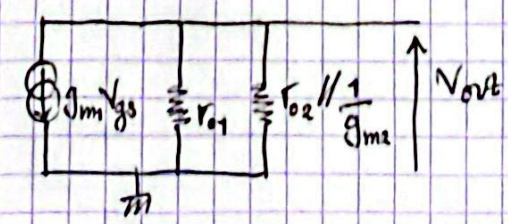
$A_v = -g_m(r_{o1} // r_{o2})$

③



$R_{out} = r_{o1} // (\frac{1}{g_{m2}} // r_{o2})$

Gain



$A_v = \frac{V_{out}}{V_{in}}$ $\quad V_{in} = V_{gs}$

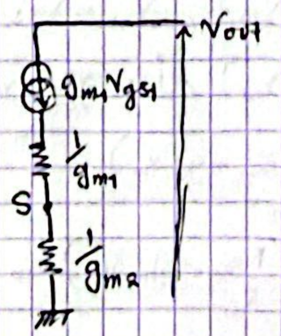
$V_{out} = -g_m V_{gs} * R_{out}$

$= -\frac{g_m V_{gs} R_{out}}{V_{gs}}$

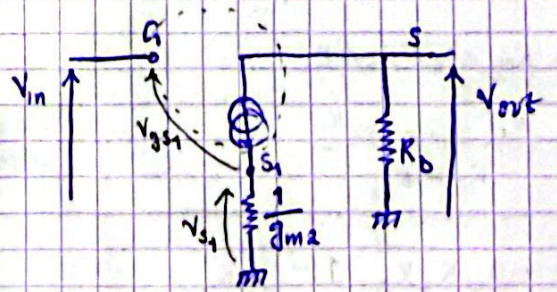
$A_v = -g_m R_{out} = -g_{m1} \cdot (r_{o1} // (\frac{1}{g_{m2}} // r_{o2}))$

EXERCICE 2

① $\lambda = 0$; Gain?



$R_{out} = \frac{1}{g_{m1}} + \frac{1}{g_{m2}}$



$A_v = \frac{V_{out}}{V_{in}}$

$* \; V_{in} = V_{gs1} + \frac{1}{g_{m2}} \cdot g_{m1} V_{gs1}$

$V_{gs1} = \frac{1}{1 + \frac{g_{m1}}{g_{m2}}} \cdot V_{in}$

$V_{out} = -R_D \cdot g_{m1} V_{gs1}$

$= \frac{-R_D \cdot g_{m1} \cdot V_{in}}{1 + \frac{g_{m1}}{g_{m2}}}$

$A_v = \frac{-g_{m1} R_D}{1 + \frac{g_{m1}}{g_{m2}}}$

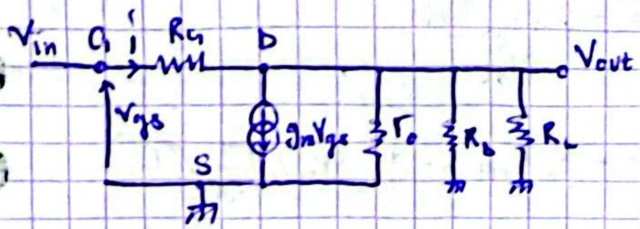
$= \frac{-R_D}{\frac{1}{g_{m1}} + \frac{1}{g_{m2}}}$

CS degenerates

$A_v = \frac{-R_D}{\frac{1}{g_{m1}} + R_S}$

TD2 : EXERCICE 7

① A_v (small-signal)



* $V_{out} = ?$

with $R_{éq} = R_L // R_D // r_o$

$$V_{out} = R_{éq}(i - g_m V_{gs}) \quad , \quad V_{in} = V_{gs}$$

$$= R_{éq}(i - g_m V_{in}) \quad , \quad i = \frac{V_{in} - V_{out}}{R_G}$$

$$= R_{éq}\left(\frac{V_{in} - V_{out}}{R_G} - g_m V_{in}\right)$$

$$V_{out}\left(1 + \frac{R_{éq}}{R_G}\right) = \left(\frac{R_{éq}}{R_G} - g_m R_{éq}\right) V_{in}$$

$$\frac{V_{out}}{V_{in}} = A_v = \frac{\frac{R_{éq}}{R_G} - g_m R_{éq}}{1 + \frac{R_{éq}}{R_G}}$$

with $R_{eq} = r_o // R_D // R_L$

$$A_v = \frac{R_{éq} - g_m R_{eq} R_G}{R_G + R_{éq}}$$

$$= -g_m R_L R_D \cdot \frac{1 - \frac{1}{g_m R_G}}{1 + \frac{R_L R_D}{R_G}}$$

si $g_m R_G >> 1$

$$A_v \approx -g_m R_L R_D$$

$$A_v = \frac{R_{éq} - g_m R_{éq} R_G}{R_G + R_{éq}}$$

$$= R_{éq} \frac{1 - g_m R_{éq}}{R_G + R_{éq}}$$

$$= -g_m R_{éq} \frac{\frac{-1}{g_m} + R_{éq}}{R_G + R_{éq}}$$

* si $g_m R_D >> 1$

$$A = -g_m R_{éq}$$

$$R'_{in} = \frac{V_{in}}{i}$$

$$i = \frac{V_{in} - V_{out}}{R_G} \quad , \quad V_{out} = -g_m R_{éq} V_{in}$$

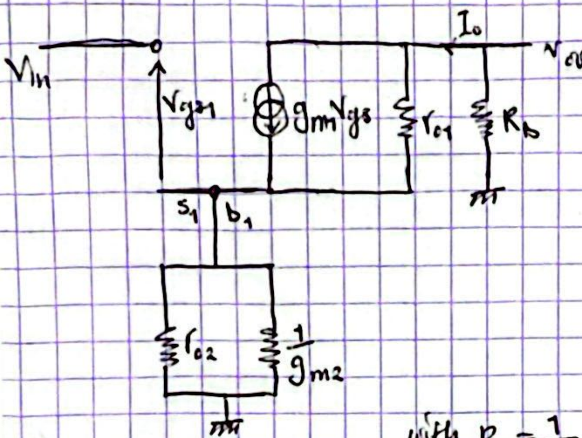
$$i = V_{in} \frac{(1 + g_m R_{éq})}{R_G}$$

$$R'_{in} = \frac{R_G}{1 + g_m R_{éq}}$$

$$R_{out} = r_o // R_D // R_L$$

EX 2

$\lambda \neq 0$



with $R_S = \frac{1}{g_{m2}} // r_{o2}$

Gain A_v?

$V_{out} = -R_D I_o$, $I_{R_S} = I_o = -\frac{V_{out}}{R_D}$

* Calculate I_{ro1}

$I_{ro1} = I_o - g_{m1}V_{gs1}$

with $V_{gs1} = V_{in} - V_{S1}$, $V_{S1} = R_S \cdot I_o$

$= \frac{-R_S}{R_D} V_{out}$

$I_{ro1} = -\frac{V_{out}}{R_D} - g_{m1}V_{gs1}$

$= \frac{-V_{out}}{R_D} - g_{m1}\left(V_{in} + \frac{R_S}{R_D}V_{out}\right)$

In eq. $V_{out} = r_{o1} I_{ro1} + V_{S1}$

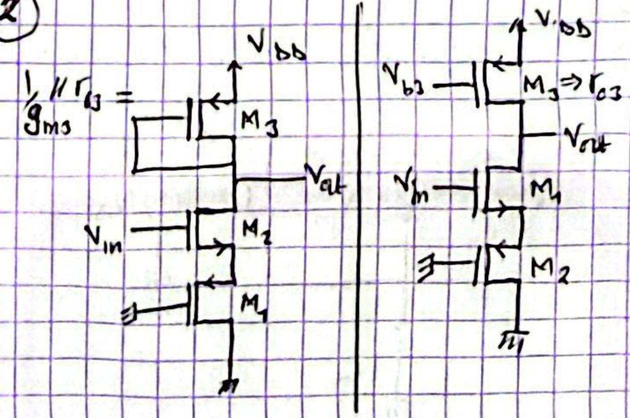
$$V_{out} = r_{o1}\left(\frac{-V_{out}}{R_D} - g_{m1}\left(V_{in}+\frac{R_S V_{out}}{R_D}\right)\right) - \frac{R_S}{R_D}V_{out}$$

$$\frac{V_{out}}{V_{in}} = \frac{-g_{m1} r_{o1} R_D}{R_D+R_S+r_{o1}+g_{m1}r_{o1}R_S}$$

si $\lambda=0$ $\frac{V_{out}}{V_{in}} = \frac{r_{o1}(-g_{m1}R_D)}{r_{o1}\left(1+\frac{R_D}{r_{o1}}+\frac{R_S}{r_{o1}}+g_{m1}R_S\right)}$

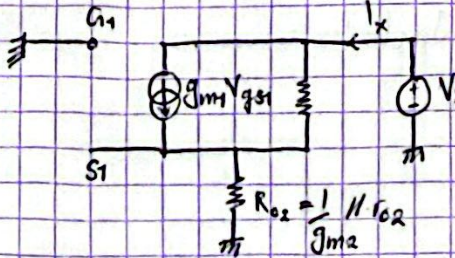
$= \frac{-g_{m1}R_D}{1+g_{m1}R_S}$

(2)



EXERCICE 3

(1)



with $R_{out} = \frac{V_x}{I_x}$

$V_{gs1} = V_{g1} - V_{S1} = -V_{S1} = -R_{o2} I_x$

* Current $I_{ro1} = I_x - g_{m1}V_{gs1} = I_x + g_{m1}R_{o2}I_x$

* $V_x = r_{o1} \cdot I_{ro1} + R_{o2} \cdot I_x$

$= (r_{o1}(1+g_{m1}R_{o2}) + R_{o2}) I_x$

$R_{out} = \frac{V_x}{I_x} = r_{o1} + R_{o2} + g_{m1} r_{o1} R_{o2}$

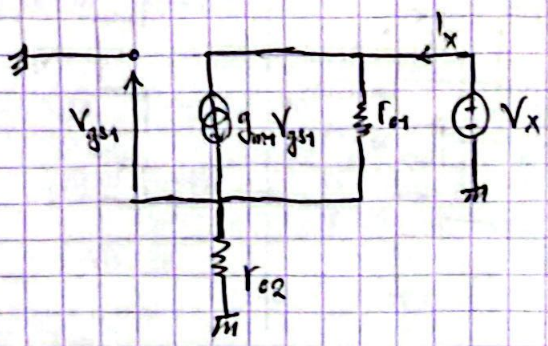
with $R_{o2} = \frac{1}{g_{m2}} // r_{o2} = \frac{r_{o2}\cdot\frac{1}{g_{m2}}}{r_{o2}+\frac{1}{g_{m2}}} = \frac{r_{o2}}{1+g_{m2}r_{o2}}$

$R_{out} = r_{o1} + \frac{1}{g_{m2}} // r_{o2} + g_{m1} \cdot r_{o1} \cdot \left(\frac{1}{g_{m2}} // r_{o2}\right)$

* si $\lambda_{1,2} = 0$

$R_{out} = r_{o1} + \frac{1}{g_{m2}} + \frac{g_{m1} r_{o1}}{g_{m2}}$

②



with analog in question (1)

$R_{out} = r_{o1} + r_{o2} + g_m r_{o1} r_{o2}$

$R_{out} \simeq g_{m1} r_{o1} r_{o2}$